

Ledao Chen

Senior AI & ML Engineer | Data Scientist

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SUMMARY

Senior AI/ML Engineer with 8 years of experience building scalable ML systems, GenAI solutions, and cloud-native data pipelines across gaming, consumer tech, and financial data industries. I focus on practical problem-solving, clean engineering, and applying advanced LLM and unsupervised learning methods to real business problems. Known for blending deep modeling intuition with strong production engineering skills to deliver systems that are reliable, interpretable, and easy for teams to adopt.

SKILLS HIGHLIGHT

- **Cloud & MLOps:** GCP (BigQuery, Vertex AI, Dataflow, Cloud Functions, Cloud Storage, Looker Studio), AWS (S3, Lambda, Glue, Redshift, SageMaker), Docker, Kubernetes, CI/CD, MLflow, Airflow
- **Machine Learning:** Unsupervised ML, Self-supervised learning, Transformers, RAG, Clustering, Embeddings, Graph ML, Foundation Models, Deep Learning, Sequence Modeling
- **GenAI & LLM Tools:** OpenAI API, Vertex AI GenAI Studio, Hugging Face Transformers, LangChain, LlamaIndex, StabilityAI, DALL·E, RunwayML, Diffusion Models, Prompt Engineering
- **Data Engineering:** SQL, BigQuery ML, Dataflow pipelines, ETL/ELT design, Feature stores, Streaming pipelines, API integration, Microservices
- **Programming:** Python, PyTorch, TensorFlow, scikit-learn, Java, Shell, FastAPI, Flask
- **Visualization & Storytelling:** Plotly, Tableau, Looker Studio, Kibana, Matplotlib, generative visual storytelling workflows
- **Other:** Statistics, Data modeling, Feature engineering, Model introspection, Knowledge graphs, Ontologies, Benchmarking, Distributed systems

PROFESSIONAL EXPERIENCE

PlayStation

Remote

Senior AI & ML Engineer | Data Scientist

Sep 2022 – Current

- Led development of a scalable unsupervised learning framework using PyTorch and Vertex-AI-aligned abstractions to analyze gameplay telemetry, producing robust clustering and embedding structures that informed personalization and downstream GenAI planning behaviors in production ML systems.
- Designed generative AI pipelines for automated content summarization and visual storytelling of player sessions using OpenAI API, Hugging Face Transformers, and multimodal diffusion models, enabling creative teams to rapidly explore gameplay dynamics through narratively coherent summaries and imagery.
- Partnered with full-stack engineering teams to integrate ML microservices into PlayStation platform features using Python, FastAPI, and GCP-oriented deployment practices such as Cloud Functions and containerized inference endpoints mapped from internal orchestration systems.
- Built a distributed data ingestion and modeling pipeline inspired by Dataflow patterns to process billions of gameplay events, ensuring real-time feature extraction and enabling faster experimentation with self-supervised representation models and graph-based player behavior analysis.
- Developed RAG prototypes to support internal analytics teams, leveraging LangChain, LlamaIndex, and BigQuery-style retrieval constructs to enable natural-language querying over massive historical gameplay repositories with strong latency guarantees.
- Implemented domain-specific prompt engineering strategies, including chain-of-thought, reflection, and prompt chaining mechanics, to improve LLM reasoning over player logs, telemetry sequences, and content metadata, boosting model interpretability and stability.

- Created knowledge graph structures describing interactions between players, in-game mechanics, and narrative elements, enabling search-driven insights and agent-planning workflows that mirror Clear Fracture’s approach to domain-specific ontologies.
- Performed deep model introspection on transformer-based sequences for event prediction tasks, documenting attention patterns, activation distributions, and failure modes to guide cross-team design decisions and improve model transparency.
- Collaborated with product and client-facing internal groups to create compelling visualizations using Plotly, Tableau, and generative visualization tools like RunwayML, enabling stakeholder-level storytelling on engagement and game economy trends.
- Led investigations into emerging multimodal foundation models and orchestrated proof-of-concept studies comparing model architectures, evaluating cost-performance tradeoffs, and defining benchmarks that aligned with broader platform strategy and agent-centric pipelines.

Apple

Austin, TX

Senior AI & ML Engineer | Data Scientist

Jul 2021 – Sep 2022

- Architected semi-supervised and unsupervised behavioral models using TensorFlow and scikit-learn to extract high-level usage insights from large unstructured mobile telemetry datasets, building structured representations used by downstream personalization and information-retrieval systems.
- Designed generative AI workflows that automated summarization of user behavior segments using OpenAI and Hugging Face Transformers, enabling product teams to evaluate feature interactions through concise narratives aligned with visual plots and dynamic dashboards.
- Implemented scalable data pipelines on AWS resembling GCP Dataflow-style orchestration, allowing a later migration path toward BigQuery and Vertex AI conventions; this included event federation, transformation layers, and efficient embedding storage.
- Collaborated with full-stack teams to deploy ML-powered APIs and microservices, adopting similar abstraction patterns as Cloud Functions and Vertex AI endpoints for rapid experimentation and controlled, monitored rollout.
- Introduced RAG techniques for internal knowledge-surfacing tasks using early LangChain patterns and custom embedding stores, providing a proof-of-concept for semantic search across product logs, developer documentation, and behavioral studies.
- Created interpretability modules for transformer-based models using attention inspection, neuron activation monitoring, and attribution methods, improving trust for analytics teams and informing strategic recommendations for user experience improvements.
- Developed interactive visual storytelling tools using Plotly and Tableau to help non-technical stakeholders understand segmentation outcomes, latent pattern discovery, and user-journey transitions produced through unsupervised modeling.
- Led iterative prompt-engineering experiments—reflection, reasoning-and-act (ReAct), and prompt chaining—to optimize LLM-based recommendation prototypes, evaluating multi-agent decision processes and their alignment with product constraints.
- Drove cross-functional design sessions to define data models, conceptual diagrams, and system documentation for ML integrations, ensuring internal teams could test, validate, and benchmark models using consistent standards.
- Evaluated new multimodal and agentic model architectures, performed alternatives analyses, and contributed to technical roadmaps describing how emerging LLM-adjacent technologies could enhance large-scale consumer data understanding.

Equifax

Atlanta, Georgia

Data Scientist

Jul 2018 – Jul 2021

- Developed unsupervised ML models in Python and scikit-learn to detect credit behavior anomalies and latent customer clusters using structured financial datasets, establishing feature-engineering pipelines that later informed graph-based relationship analysis.
- Built early ETL workflows using SQL and distributed processing principles similar to modern Dataflow patterns, enabling reliable ingestion of credit bureau data and supporting downstream modeling tasks with improved data latency and consistency.
- Created internal visualization dashboards with Tableau and Matplotlib to illustrate risk distributions, model drift, and population stability metrics, enhancing collaboration with business, compliance, and risk-management teams.
- Implemented semi-automated model introspection functions using custom statistical diagnostics, documenting model assumptions, sensitivities, and feature dependencies to meet rigorous validation requirements.
- Collaborated with engineering teams to integrate predictive models into operational environments via Python APIs and batch scoring workflows, ensuring reliable performance and strong alignment with compliance constraints.
- Contributed to early experiments involving embeddings and graph-based ML to analyze relationships between credit entities, laying groundwork for more modern self-supervised methods used in later roles.
- Designed statistical process flow diagrams, data models, and testing plans for regulated scoring models, ensuring clarity, reproducibility, and compliance documentation for audit reviews.
- Worked closely with stakeholders to communicate findings clearly through presentations, emphasizing data storytelling and narrative clarity that later informed more sophisticated GenAI-driven visualization approaches.

Equifax

Atlanta, Georgia

Model Validation Intern

May 2018 – Aug 2018

- Supported validation of supervised and unsupervised credit risk models using statistical diagnostics, sensitivity testing, and population drift analysis, building a strong foundation in rigorous model assurance practices relevant for modern ML safety and interpretability.
- Conducted feature-stability evaluations using SQL and Python, ensuring data integrity and documenting issues that were escalated to modeling teams for correction or retraining.
- Generated detailed model documentation, validation write-ups, and reproducibility reports, demonstrating an ability to translate complex methodological details into clear, stakeholder-friendly language.
- Implemented data-processing scripts and reproducibility notebooks that mirrored modern pipeline concepts, improving consistency across validation cycles and reducing manual workload.
- Analyzed historical model failures to identify root causes, informing stronger testing frameworks and helping establish structured evaluation templates used by senior validators.
- Developed visual summaries of validation outcomes using Matplotlib and Tableau, contributing to clearer communication of risk scoring performance and operational stability.
- Participated in multi-team review sessions to understand model intent, assumptions, and business alignment, building early experience with cross-functional collaboration.
- Reviewed documentation for compliance and regulatory alignment, gaining familiarity with governance frameworks that reinforce responsible ML deployment today.

EDUCATION

University of Virginia

Master of Science in Statistics

2016 - 2018

Charlottesville, VA

Sichuan University

Bachelor of Science in Applied Mathematics

2012 - 2016

Si Chuan Sheng, China